

Bug Report

1. Silent Failure on Missing Output Directory

File: `Main.java` (lines 17-21, 212-217) **Description:** The application tries to clear and append to `output/sample_outputs.txt`. If the `output/` directory does not exist, a `FileNotFoundException` is caught and ignored silently. The user is not notified, and receipts are never saved. **Severity:** Medium **Fix:** Add `new File("output").mkdirs();` before writing to ensure the directory exists.

2. Null Pointer Exception Risk in Cart Operations

File: `Cart.java` (line 14) **Description:** The `addItem` method assumes the `product` argument is non-null. If `product` is null, calling `product.getStockQuantity()` throws a `NullPointerException` rather than throwing a handled `IllegalArgumentException`. **Severity:** Low

3. Redundant and Inefficient Total Calculations

File: `Cart.java` (lines 69-96) **Description:** Calling `calculateTotal()` internally calls `calculateTax()`, `calculateDiscount()`, and `calculateSubtotal()`. `calculateTax()` also calls `calculateSubtotal()` and `calculateDiscount()`. This results in `calculateSubtotal()` being executed multiple times, which iterates through all items in the cart each time. **Severity:** Low (Performance)

4. Default Charset Usage in File I/O

File: `FileHandler.java` **Description:** `FileReader` and `FileWriter` use the system's default character encoding. Since the codebase uses the Rupee symbol (₹), this can cause mojibake (garbled text) when run on systems with different default encodings (e.g., Windows CP-1252 instead of UTF-8). **Severity:** Medium **Fix:** Specify UTF-8 explicitly: `new InputStreamReader(new FileInputStream(filePath), StandardCharsets.UTF_8)`.

5. Potential Negative Discount Edge Case

File: `Coupon.java` (line 28) **Description:** If a `FLAT` discount value exceeds the subtotal, `Math.max(0, subtotal - discountValue)` correctly ensures the discounted amount doesn't go below 0. However, in `Cart.java` line 83, `subtotal - afterDiscount` calculates the discount amount. This logic holds up, but checking for negative or \$0 carts should be properly handled to avoid 0 cart checkouts if that is a business rule. **Severity:** Low